Huawei Access Node HEB-ACC-33 QoS Congestion Event Troubleshooting and Resolution SOP

Node: HEB-ACC-33 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_6655

Description

The occurrence of QoS congestion events on Huawei Access Node HEB-ACC-33 can be attributed to the complex interplay between the QoS scheduling algorithm and the physical interface limitations. The Huawei Access Node employs a hierarchical QoS architecture, where traffic classes are mapped to queues, and these queues are serviced by a hierarchical scheduler. When microbursts or interface tail drops occur, it can lead to QoS queue starvation, resulting in a congested interface. This congestion can propagate upstream, affecting the overall network performance. The severity of this issue is rated as minor, but it is essential to address the root cause to prevent potential degradation of services. The following SOP outlines the procedures to diagnose and resolve QoS congestion events on Huawei Access Node HEB-ACC-33.

Pre-requisite Checks

- check: Verify that the redundant link status is active and operational at Hebbal Edge., description: Ensure that the redundant link is functioning correctly to prevent any potential impact on the network.

- check: Confirm that the Huawei Access Node is running the latest software version., description: Ensure that the node is running the latest software to prevent any known issues or bugs.

- check: Perform a thorough physical inspection of the node and its connections., description: Identify any potential physical issues that may be contributing to the congestion event.

Potential Root Causes

Cause	Probability
QoS scheduling algorithm misconfiguration	High
Interface physical limitations (e.g., oversubscription)	Medium
Software bugs or defects	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos queue statistics, Expected Output: Queue statistics, including queue depth, packet drop count, and packet loss rate., Description: This command will provide insight into the queue depth, packet drop count, and packet loss rate, helping to identify potential queue starvation conditions.

Command: display interface counters, Expected Output: Interface counters, including packet count, byte count, and error count., Description: This command will provide insight into the interface traffic patterns, helping to identify potential issues with the
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interface or its connections.

Command: display qos hierarchical scheduler, Expected Output: QoS
hierarchical scheduler configuration and statistics., Description:
This command will provide insight into the QoS scheduling algorithm
configuration and statistics, helping to identify potential
misconfigurations or issues with the scheduler.
```

Resolution Steps

Step 1: Step: Adjust QoS scheduling algorithm configuration, Command: set qos queue scheduling-algorithm , Description: Adjust the QoS scheduling algorithm configuration to optimize queue servicing and prevent queue starvation conditions.

Step 2: Step: Modify interface settings to address physical limitations, Command: set interface bandwidth , Description: Modify the interface settings to address physical limitations, such as oversubscription, and prevent congestion events.

Step 3: Step: Upgrade software to latest version, Command: system software upgrade, Description: Upgrade the software to the latest version to address any known issues or bugs that may be contributing to the congestion event.

Post-Check Verification

```
Command: display qos queue statistics, Expected Output: Queue
statistics showing no signs of queue starvation or packet loss.,
Description: Verify that the queue statistics show no signs of queue
starvation or packet loss to confirm the fix.

Command: display interface counters, Expected Output: Interface
counters showing normal traffic patterns., Description: Verify that
the interface counters show normal traffic patterns to confirm the
fix.

Command: display qos hierarchical scheduler, Expected Output: QoS
hierarchical scheduler configuration and statistics showing optimal
performance., Description: Verify that the QoS scheduling algorithm
configuration and statistics show optimal performance to confirm the
fix.
```

Roll-back Procedures

- Step: Restore original QoS scheduling algorithm configuration, Command: set qos queue scheduling-algorithm , Description: Restore the original QoS scheduling algorithm configuration to its previous state in case the fix was unsuccessful.

- Step: Revert interface settings to original configuration, Command: set interface bandwidth , Description: Revert the interface settings to their original configuration in case the fix was unsuccessful.

- Step: Revert software to previous version, Command: system software rollback, Description: Revert the software to its previous version in case the fix was unsuccessful.